HW 6:

Problem 1:

$$V_0 = \frac{R_2}{R_1} V_{id}$$

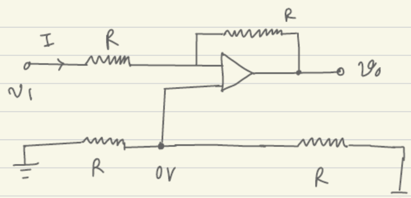
$$R_{id} = 10K\Omega$$

$$A_{cm} = 0.0095$$

$$CMRR = 66.4dB$$

Problem 2:

Prob 2:



$$v_1 + \frac{v_0 - v_1}{2} = \frac{v_2}{2}$$

or, $v_0 = v_2 - v_1$

v_1 only : $R_i = \frac{v_1}{I} = R$

v_2 only: $R_i = \frac{v_2}{I} = 2R$

V between 2 terminals: $R_i = v/I = 2R$

v_1 connected to both input terminals:

$I_1 = V_1/2R + V_1/2R = V_1/R$

$R_i = R$

Problem 3:

Prove the equation

Problem 4:

$$A_d = 1$$

$$A_{cm} = 0$$

$$-5V \le V_{cm} \le 5V$$

$$A_{cm} = 0$$

$$A_d = 10$$

$$-3V \le V_{cm} \le 3V$$

Problem 5:

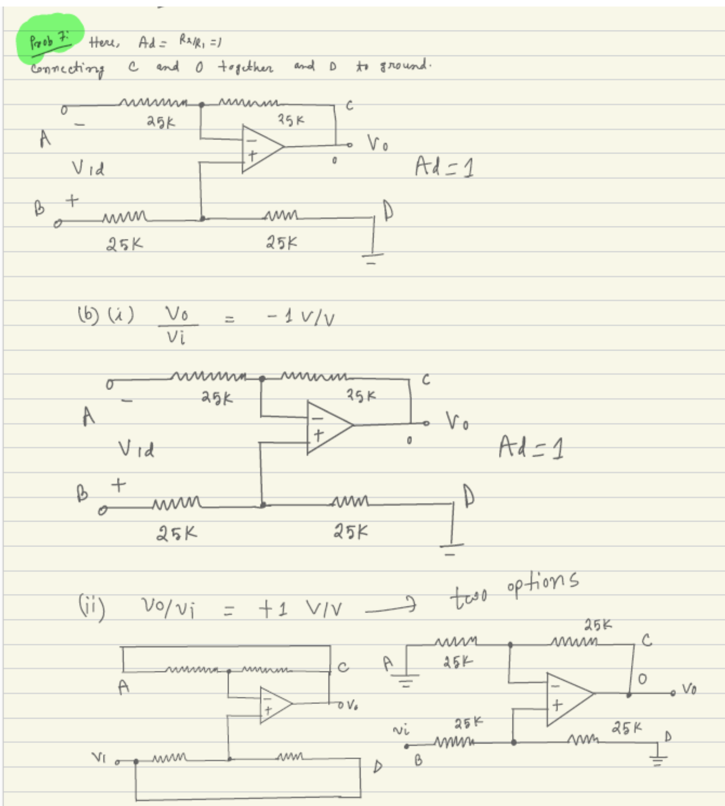
$R_5 = 756\Omega$

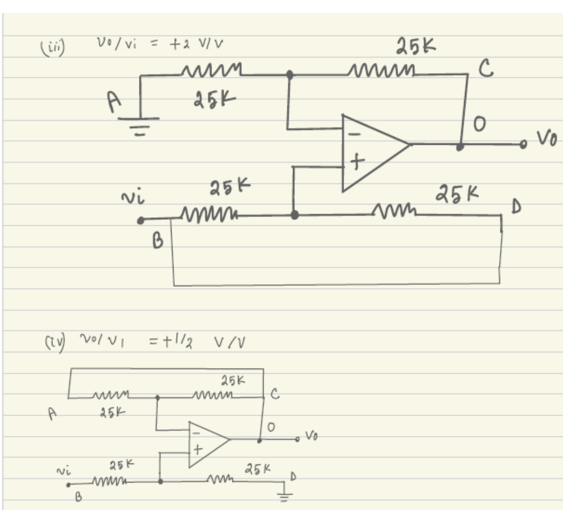
$R_6 = 6.8K\Omega$

Problem 6:

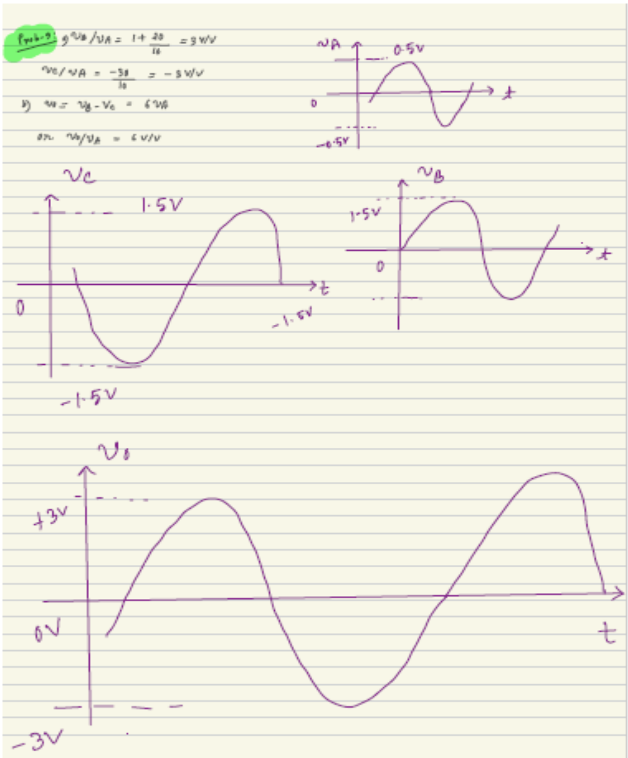
Prove

Problem 7:





Problem 9:



$$-28V \le V_0 \le 28V$$

$$56V = V_{p-p}$$

$$V_{r.m.s} = 19.8V$$

Problem 10:

Largest V that can be applied = 14mV

V_rms=100mV

Problem 11:

V_0=-10V

V0=4V

Vp=40mV

R_L_min=502 Ohm

Problem 13:

$$\frac{V_{out}}{V_{in}} = 4.66$$
$$I_{in} = 0A$$
$$R_{in} = \inf$$
$$V = 30V$$

Problem 14:

Vout=9.05 V